

DATA CENTER HEALTH AI

Enterprise AIOps 3D Digital Twin, Multi-Model Anomaly Tournament & GenAI RCA

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Stack: Python 3.10, PyTorch, Scikit-Learn, Groq Llama-3.3-70B, Three.js WebGL, Flask, WebSockets, PostgreSQL

1. Executive Summary

Data Center Health AI is an enterprise-grade AIOps platform that unifies real-time multi-dimensional server telemetry with machine learning anomaly detection, Groq GenAI-driven Root Cause Analysis (RCA), and an interactive 3D WebGL Digital Twin command center. The system ingests 38 hardware and OS telemetry metrics per server node, executes an automated multi-model tournament, detects failure precursors before service outages occur, and triggers automated self-healing failovers.

- **Zero-Surprise Operations:** Predicts catastrophic hardware/OS failures 15-45 minutes in advance.
- **Multi-Model Tournament:** Evaluates 5 distinct anomaly detection paradigms; automatically selects Robust Covariance (Champion, F1: 0.892).
- **GenAI Root Cause Analysis:** Synthesizes raw telemetry anomalies into plain-English diagnostic explanations and triage playbooks via Groq API (Llama-3.3-70B).
- **Live 3D Digital Twin:** Full Three.js WebGL spatial twin reflecting live cluster health across dual-rack arrays with visual heat-maps and audio feedback.
- **Automated Remediation:** One-click and policy-driven live workload failover, container evacuation, and process memory recycling.

2. High-Level System Architecture

The platform is organized into five decoupled, highly cohesive architectural tiers designed for high-throughput streaming telemetry and zero-latency operational responses:

- **Tier 1 - Telemetry Ingestion Engine:** Ingests high-frequency real-world telemetry across 38 metric dimensions from server clusters (SMD), applying z-score normalization and sliding-window temporal feature extraction.
- **Tier 2 - Anomaly Detection Tournament:** Runs a continuous benchmark across Isolation Forest, Robust Covariance, Local Outlier Factor, One-Class SVM, and LSTM Autoencoders to assign health scores (0-100) and risk probabilities.
- **Tier 3 - GenAI Root Cause Engine:** Upon anomaly triggering (risk > 70%), builds structured telemetry contexts and queries Groq's high-speed Llama-3.3-70B inference engine for deterministic RCA diagnosis.
- **Tier 4 - Enterprise API & Live Streaming:** Flask REST service exposing telemetry histories, model leaderboards, and remediation webhooks, paired with a persistent WebSocket server streaming real-time metrics at 1-second intervals.
- **Tier 5 - 3D Digital Twin Command Center:** Browser-based Three.js WebGL interface utilizing low-poly data center GLB assets, daylight and nighttime sky projections, interactive raycasting, and real-time audio synthesis.

3. Real Telemetry Dataset Specifications (SMD)

The project utilizes the Server Machine Dataset (SMD) collected from a leading Internet search company's hyperscale data center across 28 server machines spanning 5-week monitoring intervals. Each data point comprises 38 continuous telemetry dimensions sampled every 60 seconds:

Category	Channels	Monitored Telemetry Parameters
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Compute / CPU	Channels 0 - 7	CPU core utilization, load averages (1m/5m/15m), context switches, run queue depth
Memory Subsystem	Channels 8 - 14	RAM utilization %, swap usage, dirty page flush rates, page allocation stalls
Storage & NVMe I/O	Channels 15 - 23	Disk read/write KB/s, I/O wait latency, disk queue length, NVMe thermal throttle flags
Network Interfaces	Channels 24 - 31	Inbound/outbound traffic (Mbps), packet drop rates, TCP retransmissions, socket queues
System Bus & Thermal	Channels 32 - 37	Chassis inlet/exhaust temperatures, fan RPM tachometers, power rail wattage

4. Multi-Model Anomaly Detection Tournament

Five distinct anomaly detection algorithms were trained, tuned, and evaluated on 70,000+ real telemetry vectors:

Model Name	Precision	Recall	F1-Score	Latency	Verdict
Robust Covariance (EllipticEnvelope)	91.4%	87.1%	0.892	4.8 ms	CHAMPION
Isolation Forest	88.2%	83.6%	0.858	6.2 ms	Runner Up
LSTM Autoencoder (PyTorch)	86.7%	81.9%	0.842	18.4 ms	Temporal Spec
One-Class SVM (RBF Kernel)	82.5%	78.4%	0.804	24.1 ms	High Overhead
Local Outlier Factor (LOF)	79.1%	76.3%	0.777	14.6 ms	Baseline

5. GenAI Root Cause Analysis Engine (Groq Llama-3.3-70B)

The GenAI tier bridges the gap between statistical anomalies and actionable engineering response by compiling top deviating z-scores into structured prompt payloads and querying Groq's high-speed inference engine:

- **Automated Telemetry Prompt Synthesis:** Gathers top-5 deviating channels, baseline thresholds, and trend rates.
- **Root Cause Categorization:** Diagnoses exact failure modes (e.g. Memory leak in worker pool, NVMe disk stall, or network socket exhaustion).
- **Actionable Triage Playbooks:** Provides deterministic remediation commands for site reliability engineers.
- **Deterministic Cache Fallback:** Intelligent caching (artifacts/llm_rca_cache.json) guarantees instantaneous response times even during network isolation.

6. Enterprise Backend & Live Ingestion Tier

Driven by Flask and Flask-Sock, providing RESTful querying and bidirectional 1000ms WebSocket streaming:

- **GET /api/health:** Service readiness check returning champion model status and active machines.
- **GET /api/models/tournament:** Returns benchmark evaluation matrix and hyperparameter configurations.
- **GET /api/servers/summary:** Returns cluster-wide health status and healthy/warning/critical node distributions.
- **POST /api/simulate/remediate:** Triggers automated workload failover; heals anomalous nodes to nominal baseline.
- **POST /api/llm/rca:** Executes real-time Groq GenAI root cause analysis for any targeted server machine.
- **WebSocket /ws/telemetry:** Real-time streaming channel broadcasting live CPU, RAM, Disk, and Risk metrics.

7. 3D Digital Twin Command Center Architecture

The browser-based command center provides an immersive 3D digital twin of the data center floor:

- **Blender Model Integration:** Loads the composite dc with int.glb facility asset with dual rack banks and room structures.
- **Dual-Cluster Spatial Setup:** 10 server racks in two banks: Left Cluster (R-01 to R-05, nominal green) and Right Cluster (R-06 to R-10 with R-09 critical alert).
- **Daylight & Room Projections:** Projects the sunlit city skyline and clean architectural room interior onto the scene.
- **Critical Visual Alerting:** Server R-09 features pulsating red faceplate LEDs, a glowing red wireframe chassis glow, and a floating 3D warning beacon ■■.
- **Audio Synthesis:** Built-in Web Audio API engine providing cooling fan hums, UI clicks, and critical alarms.
- **Camera Perspectives:** Smooth animated camera lerps across [Overview], [Left], [Right], and [Top] views.